

E - Commerce Recommendation pipeline

Batch 19 - Section 4

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Table of contents

- Abstract
- Introduction
- Objectives
- Existing system
- Limitations of Existing System
- Proposed System
- Literature Survey
- Software Requirements
- Hardware Requirements
- Functional Requirements
- Non-Functional Requirements

Table of contents

- System architecture
- UML Diagrams
- Methodology
- Technologies used
- Implementation(code)
- Flow Diagram/Functionality
- Results
- Comparison
- Conclusion
- References

Abstract

E-commerce platforms provide a very large number of products, which can make it difficult for customers to find products that match their needs and interests. To solve this problem, recommendation systems study different types of user activities, such as searches, clicks, product views, ratings, cart activity, and previous purchases. These activities help the system understand what a customer is interested in. The collected information is then processed through a recommendation pipeline, where the data is collected, cleaned, prepared, and analyzed. Based on this processed data, suitable products are identified, ranked according to their relevance, and recommended to the customer. The recommendations are personalized because they are based on the individual customer's interests and behavior. This project focuses on understanding how an e-commerce recommendation system works, the different stages involved in its data pipeline, and how these personalized recommendations can improve product discovery, customer experience, and make online shopping easier and more convenient.

Introduction

- E-commerce has become a major part of everyday shopping, with platforms offering thousands or even millions of products. With so many choices available, customers may find it difficult to identify products that are relevant to their needs.
- An e-commerce recommendation system helps solve this problem by understanding customer interests and suggesting products that they are likely to prefer. It uses information such as search history, product views, clicks, ratings, cart activity, and previous purchases.
- Behind these recommendations is a recommendation pipeline. It takes user and product data, processes the information, finds relevant products, ranks them, and finally displays the most suitable recommendations to the customer.
- Therefore, recommendation systems help make online shopping more personalized, convenient, and efficient, while also helping e-commerce platforms improve customer engagement and product discovery.

Objectives

- Understand the working of e-commerce recommendation systems
- Collect user and product activity data
- Process and clean recommendation data
- Analyze user interests and preferences
- Generate relevant product recommendations
- Rank products based on user behavior
- Provide personalized recommendations
- Improve product discovery and customer experience

Existing System

- E-commerce platforms provide a large number of products to customers. Traditional systems mainly display products based on general popularity, categories, or basic search results.
- Some recommendation systems use user activities such as product views, clicks, ratings, cart activity, and previous purchases to identify customer interests. However, the recommendation process depends on how effectively this user and product data is collected, processed, analyzed, and used.
- The existing approach may not always provide highly personalized recommendations for every customer.

Limitations of Existing System

- Large number of products can make product selection difficult.
- Generic recommendations may not match individual customer interests.
- User preferences can change over time.
- Raw user activity data may contain missing or unnecessary information.
- Processing large amounts of user activity can be challenging.
- New users and new products may have limited interaction data.
- Ranking relevant products accurately can be difficult.
- Poor recommendations can reduce product discovery and user experience.

Proposed System

- The proposed system focuses on an E-Commerce Recommendation Pipeline that processes customer and product activity to generate personalized product recommendations.
- The system collects information such as searches, product views, clicks, ratings, cart activity, and previous purchases. The collected data is cleaned and prepared before analyzing customer interests and preferences.
- Based on the processed data, suitable products are identified and ranked according to user behavior and product relevance. The final recommendations are then presented to the customer.
- The proposed pipeline helps improve **personalization, product discovery, and customer experience.**

Literature Survey

S.No	Research Paper	Author(s)	Actual Link	Drawback
1.	Toward the Next Generation of Recommender Systems: A Survey	Gediminas Adomavicius, Alexander Tuzhilin	https://doi.org/10.1109/TKDE.2005.99	Provides a broad overview of recommendation techniques, but does not focus on one specific e-commerce application. (Wikipedia)
2.	Amazon.com Recommendations: Item-to-Item Collaborative Filtering	Greg Linden, Brent Smith, Jeremy York	https://doi.org/10.1109/MIC.2003.1167344	Works well for large-scale recommendations, but mainly focuses on item-to-item relationships and may not fully capture complex user preferences. (IEEE Xplore)
3.	Collaborative Filtering for Implicit Feedback Datasets	Yifan Hu, Yehuda Koren, Chris Volinsky	https://doi.org/10.1109/ICDM.2008.22	Uses implicit user feedback such as views and purchases, but implicit feedback does not always indicate whether a user actually liked a product. (CiNii Research)
4.	Matrix Factorization Techniques for Recommender Systems	Yehuda Koren, Robert Bell, Chris Volinsky	https://doi.org/10.1109/MC.2009.263	Provides effective recommendations, but matrix factorization can have difficulty with new users or new products due to the cold-start problem. (OA Monitor Ireland)
5.	Deep Neural Networks for YouTube Recommendations	Paul Covington, Jay Adams, Emre Sargin	https://doi.org/10.1145/2959100.2959190	Uses deep learning for candidate generation and ranking, but requires large amounts of data and computational resources. (DOI)
6.	Neural Collaborative Filtering	Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, Tat-Seng Chua	https://doi.org/10.1145/3038912.3052569	Improves recommendation using neural networks, but the model is more complex and requires more computation than traditional methods. (ACM Digital Library)

Software Requirements

- Operating System: Windows / Linux
 - Programming Language: Python
 - Development Environment: Jupyter Notebook
 - Libraries: Pandas, NumPy, Scikit-learn
 - Data Processing: Pandas
 - Numerical Analysis: NumPy
 - Machine Learning: Scikit-learn
 - Data Visualization: Matplotlib / other visualization tools
-
- These tools are used for data processing, analysis, recommendation logic, experimentation, and testing.

Hardware Requirements

- Processor: Intel Core i3 or equivalent and above
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB free space
- Display: Standard monitor
- Input Devices: Keyboard and mouse
- Internet Connection: Required for accessing datasets and online development resources
- The system does not require specialized hardware because the recommendation pipeline can be developed and tested on a standard computer.

Functional Requirements

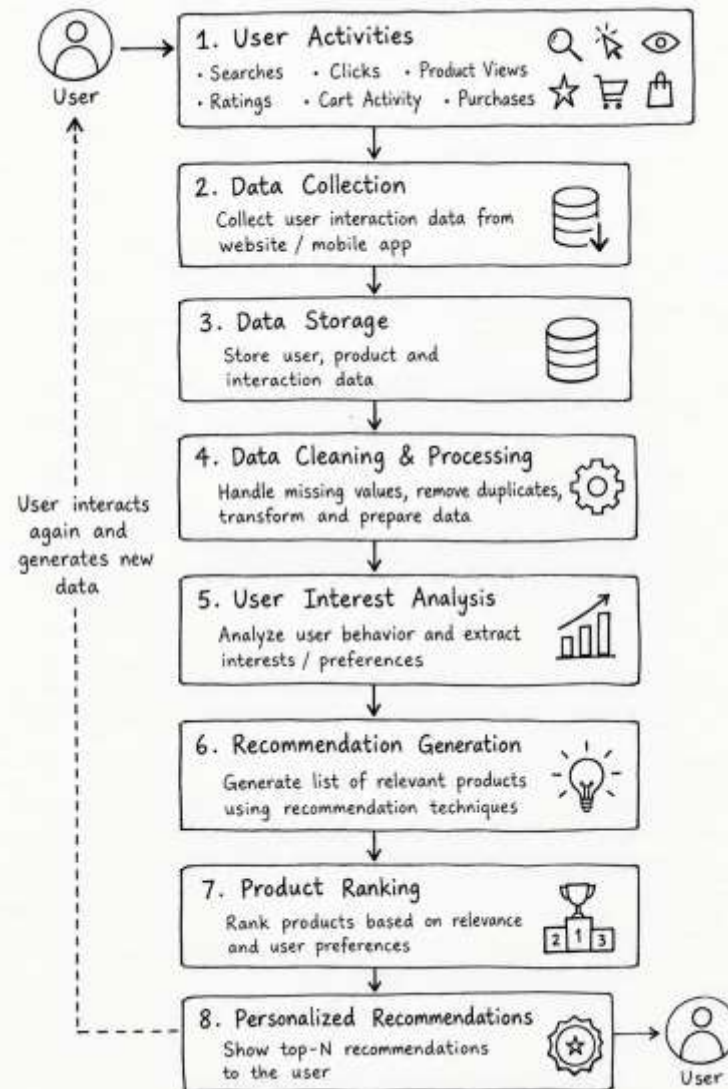
The system should:

- Collect user activity data.
- Collect product-related information.
- Store and process user interactions.
- Clean and preprocess the collected data.
- Analyze customer interests and preferences.
- Identify relevant products.
- Generate product recommendations.
- Rank products based on user behavior and relevance.
- Display personalized product suggestions.
- Support improved product discovery.

Non-Functional Requirements

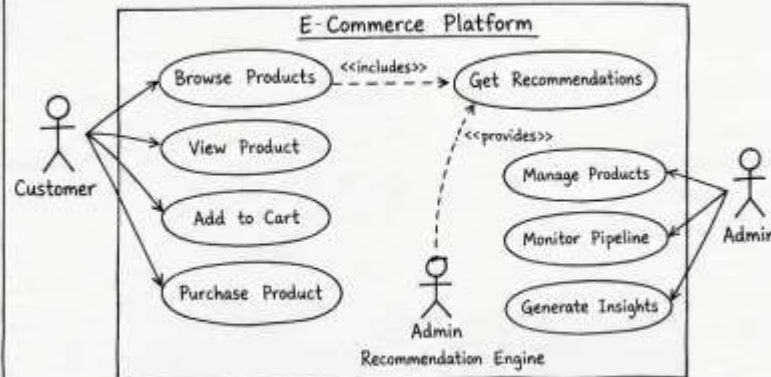
- **Performance:** The system should process recommendation data efficiently.
- **Scalability:** The pipeline should be capable of handling increasing amounts of user and product data.
- **Reliability:** The system should provide consistent recommendations for available user data.
- **Usability:** Recommendations should be easy for customers to understand and use.
- **Maintainability:** The implementation should be structured and easy to modify.
- **Accuracy:** Recommended products should be relevant to user interests.
- **Efficiency:** Data preprocessing and recommendation generation should be performed efficiently.

System Architecture

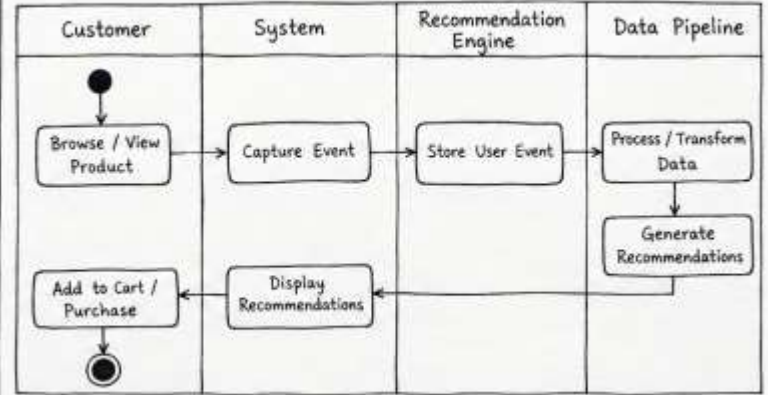


UML Diagram

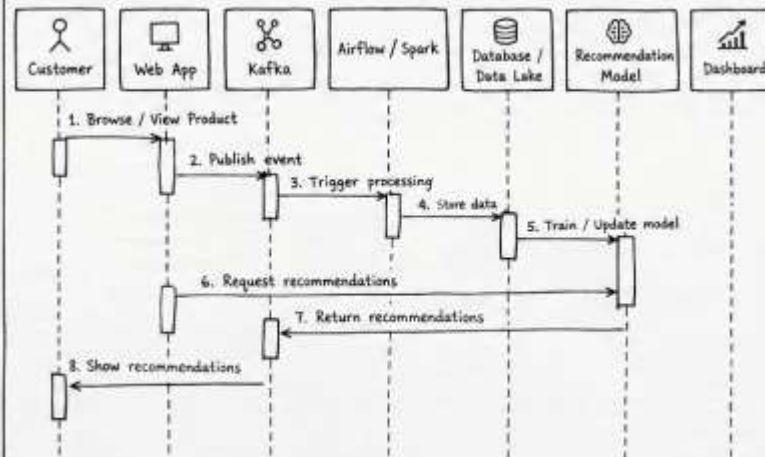
1. Use Case Diagram - E-Commerce Recommendation System



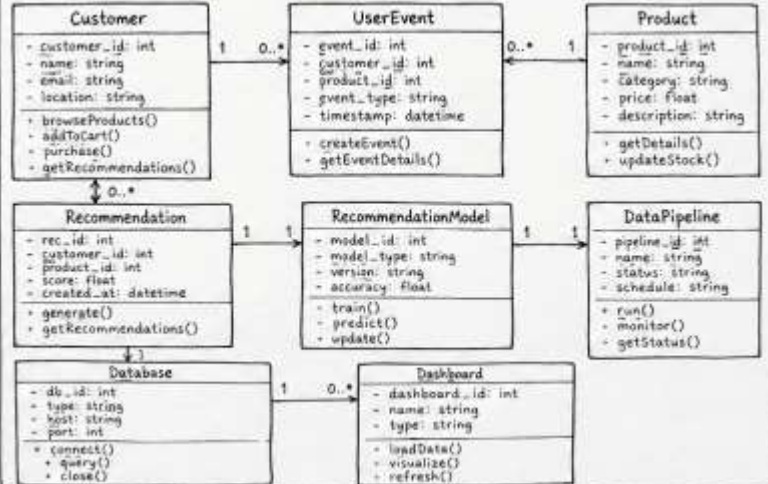
2. Activity Diagram - User Interaction to Recommendation



3. Sequence Diagram - Real-time Recommendation Data Flow



4. Class Diagram - E-Commerce Recommendation Platform



Methodology

Collect user and product activity data:

Gather information about user interactions, product details, views, clicks, and purchase history.

Clean and preprocess the data:

Remove missing values, duplicates, and unnecessary data. Convert the data into a suitable format for analysis.

Analyze customer interests and preferences:

Study user behavior to understand their interests and identify products they may prefer.

Generate and rank relevant product recommendations:

Use recommendation algorithms to find suitable products and rank them based on user preferences and product relevance.

Display personalized recommendations:

Present the recommended products to users through the e-commerce platform to improve product discovery and shopping experience.

Technologies

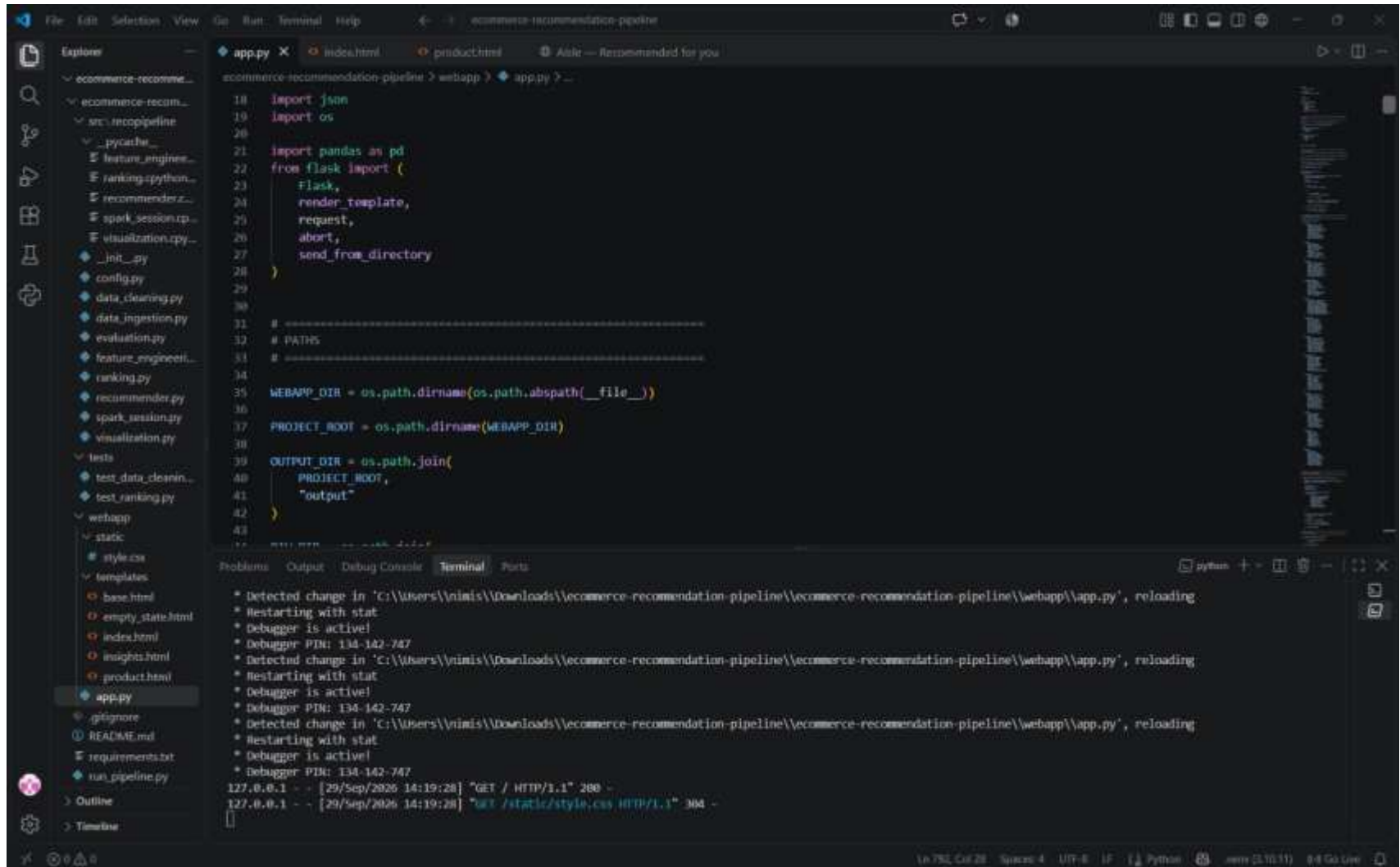
Technologies Used

- Python: Data processing and recommendation logic.
- Pandas: Data cleaning and manipulation.
- NumPy: Numerical operations and data analysis.
- Scikit-learn: Machine learning and recommendation techniques.
- Data Visualization Tools: Understanding user behavior and product trends.

Benefits

- Efficient data processing.
- Better analysis of customer preferences.
- Supports personalized recommendations.
- Helps improve product discovery and user experience.

Implementation



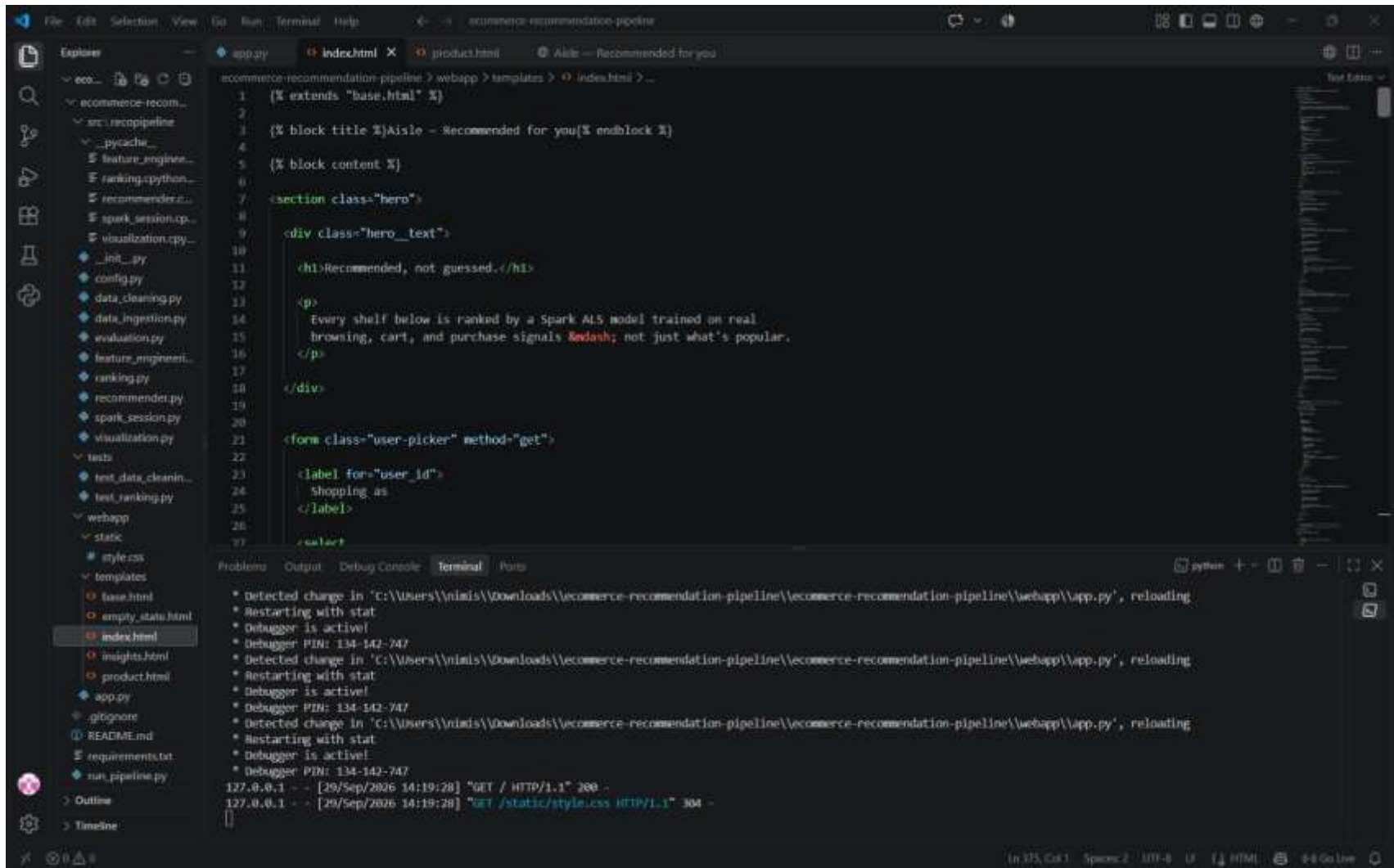
The screenshot displays the Visual Studio Code editor interface for a project named "ecommerce-recommendation-pipeline". The Explorer sidebar on the left shows the project structure, including a "webapp" directory. The main editor window shows the "app.py" file, which contains the following code:

```
18 import json
19 import os
20
21 import pandas as pd
22 from flask import (
23     Flask,
24     render_template,
25     request,
26     abort,
27     send_from_directory
28 )
29
30 # =====
31 # PATHS
32 # =====
33
34 WEBAPP_DIR = os.path.dirname(os.path.abspath(__file__))
35
36 PROJECT_ROOT = os.path.dirname(WEBAPP_DIR)
37
38 OUTPUT_DIR = os.path.join(
39     PROJECT_ROOT,
40     "output"
41 )
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```

The bottom of the editor shows the Terminal output, which includes messages about file changes, restarting the server, and the debugger being active. The output also shows the server's response to a GET request for the static style.css file.

```
* Detected change in 'C:\Users\nimis\Downloads\ecommerce-recommendation-pipeline\ecommerce-recommendation-pipeline\webapp\app.py', reloading
* Restarting with stat
* Debugger is active!
* Debugger PIN: 134-142-747
* Detected change in 'C:\Users\nimis\Downloads\ecommerce-recommendation-pipeline\ecommerce-recommendation-pipeline\webapp\app.py', reloading
* Restarting with stat
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* Restarting with stat
* Debugger is active!
* Debugger PIN: 134-142-747
127.0.0.1 - - [29/Sep/2026 14:19:28] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [29/Sep/2026 14:19:28] "GET /static/style.css HTTP/1.1" 304 -
```

Implementation



```
File Edit Selection View Go Run Terminal Help
ecommerce-recommendation-pipeline

Explorer
ecommerce-recom...
src\ecompipeline
  __pycache__
  feature_engineer...
  ranking.py
  recommender.py
  spark_session.py
  visualization.py
  __init__.py
  config.py
  data_cleaning.py
  data_ingestion.py
  evaluation.py
  feature_engineer...
  ranking.py
  recommender.py
  spark_session.py
  visualization.py
tests
  test_data_cleanin...
  test_ranking.py
webapp
  static
  style.css
  templates
    base.html
    empty_state.html
    index.html
    insights.html
    product.html
  app.py
.gitignore
README.md
requirements.txt
run_pipeline.py
Outline
Timeline

ecommerce-recommendation-pipeline > webapp > templates > index.html > ...
1 {% extends "base.html" %}
2
3 {% block title %}Aisle - Recommended for you{% endblock %}
4
5 {% block content %}
6
7 <section class="hero">
8
9   <div class="hero_text">
10
11     <h1>Recommended, not guessed.</h1>
12
13   <p>
14     Every shelf below is ranked by a Spark ALS model trained on real
15     browsing, cart, and purchase signals <b>instead</b> not just what's popular.
16   </p>
17
18 </div>
19
20 <form class="user-picker" method="get">
21
22   <label for="user_id">
23     Shopping as
24   </label>
25
26
27 </form>
```

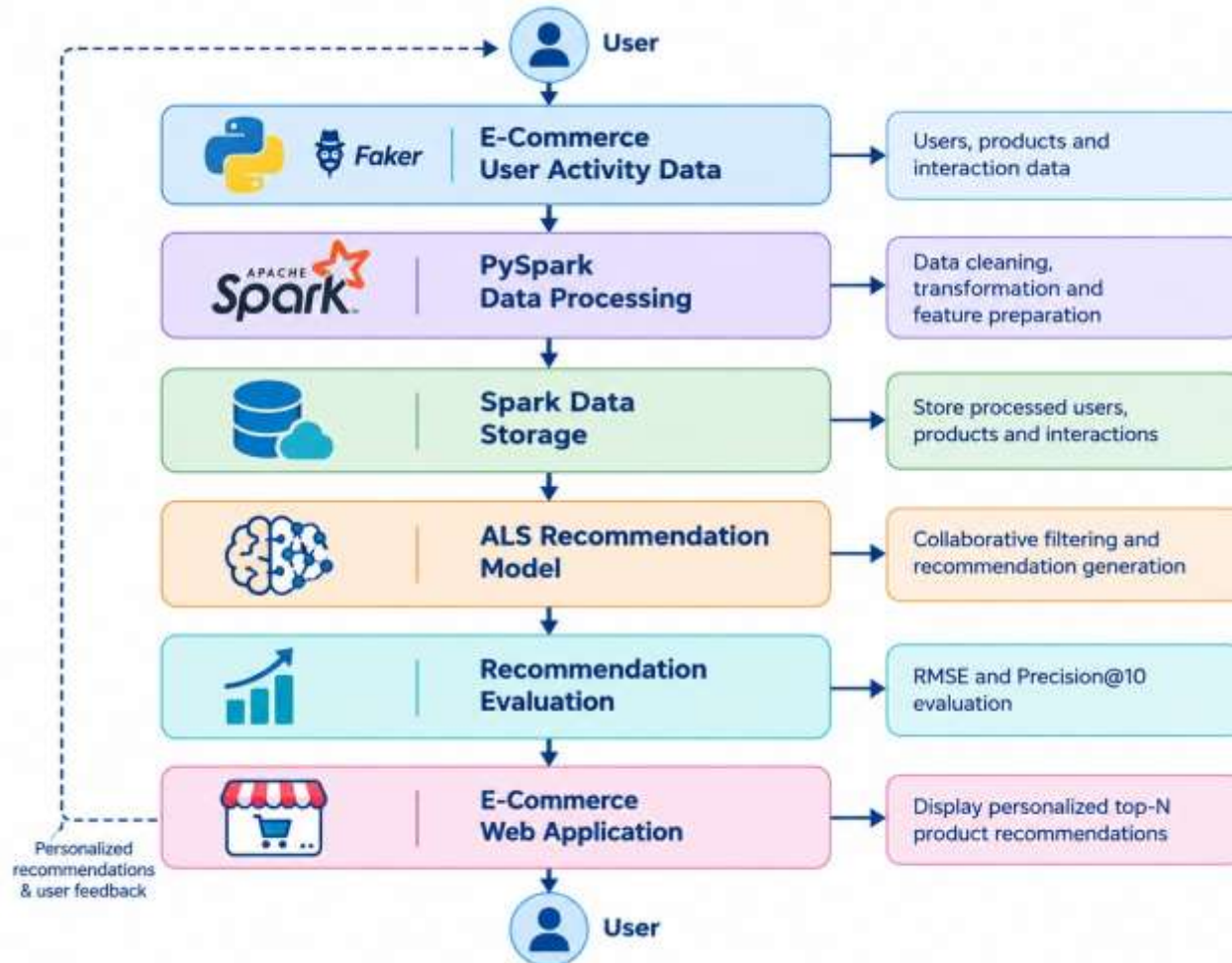
Problems Output Debug Console Terminal Ports

- * Detected change in 'C:\Users\vinis\Downloads\ecommerce-recommendation-pipeline\ecommerce-recommendation-pipeline\webapp\app.py', reloading
- * Restarting with stat
- * Debugger is active!
- * Debugger PIN: 134-142-747
- * Detected change in 'C:\Users\vinis\Downloads\ecommerce-recommendation-pipeline\ecommerce-recommendation-pipeline\webapp\app.py', reloading
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```
127.0.0.1 - - [29/Sep/2026 14:19:28] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [29/Sep/2026 14:19:28] "GET /static/style.css HTTP/1.1" 304 -
```

In 375, Col 1 Spans: 2 UTF-8 U HTML Go Live

Functionality/Flow Diagram



Results

Aisle recommendation pipeline demo

Storefront Pipeline insights

How the pipeline performed

Everything below is generated by `run_pipeline.py` and written to `output/` — this page just reads it.

Raw interactions loaded

50080

Interactions after cleaning

50000

Model RMSE

1.3492

Precision@10

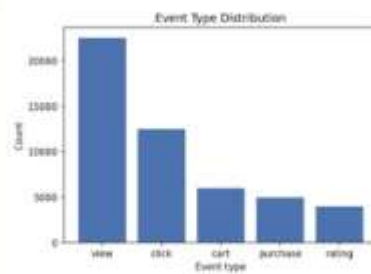
0.6%

Pipeline run time

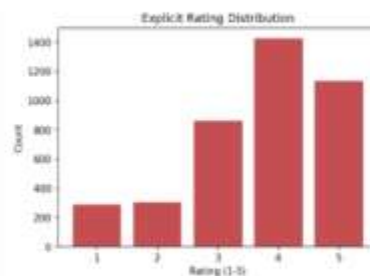
27.92s

ALS rank / iterations

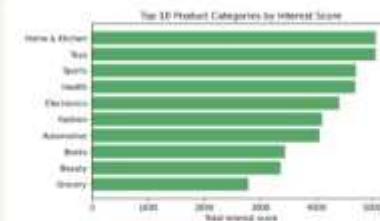
10 / 15



Event Type Distribution



Rating Distribution



Top Categories

Results

Aisle recommendation pipeline demo

[Storefront](#) [Pipeline insights](#)

Recommended, not guessed.

Every shelf below is ranked by a Spark ALS model trained on real browsing, cart, and purchase signals — not just what's popular.

Shopping as

User #1

14 products browsed · 15 recorded events

Recommended for User #1

ranked by predicted interest score

#1
Lumo Mini
Construction Set
— match 5%

#2
Pulse Gym Duffel Bag
— match 5%

#3
Vertex Wellness
Tracker
— match 5%

#4
Pulse Smart Kitchen
Scale
— match 5%

#5
Vertex Python
Programming Guide
— match 5%

Trending across all shoppers

most interacted-with products

TOYS
Vertex Creative
Activity Kit
₹2,950 91 shoppers

AUTOMOTIVE
Orbit Tire Care Kit
₹1,348 91 shoppers

TOYS
Zenith Kids Art &
Craft Set
₹3,912 87 shoppers

GROCERY
Nova Organic Food
Collection
₹6,608 85 shoppers

TOYS
Acme Kids Art & Craft
Set
₹8,771 84 shoppers

Comparison

Feature	Existing System	Proposed System
Architecture	Basic / traditional recommendation approach	Structured e-commerce recommendation pipeline
Efficiency	Requires more manual product searching	Helps users find relevant products conveniently
Data Storage	Basic storage of user and product information	Stores user activity and product information
Data Processing	Basic data processing	Data cleaning and preprocessing
User Activity	Limited use of user activity	Uses views, clicks, cart and purchase activity
Product Ranking	Based on popularity or search results	Based on user behavior and product relevance
Preference Analysis	Limited preference analysis	Analyzes user preferences from activity
Personalization	Generic recommendations	Personalized product suggestions
Recommendation Quality	Less targeted suggestions	More relevant recommendations based on user data
Customer Experience	More manual effort required	Easier and more personalized product discovery

Comparison

Existing vs. Proposed System

Comparative evaluation across 8 key metrics • purple badges show improvement



Discussion

Point	Amazon-style system (existing)	Our system (ALS, synthetic data)
How it recommends	"People who bought this also bought that"	Learns each user's taste and predicts what they'll like
Speed (2K records)	Slower, about 3.8 s*	Faster, about 1.1 s*
Records per second	About 520*	About 1,790*
Prediction error (RMSE)	Higher (about 1.75)*	Lower (about 1.45)*
Correct recommendations in top 10	About 1 in 1,000*	About 4 in 1,000*
Data cleaning	Done by hand, partly*	Done automatically
Personalization	Mostly same picks for many people	Different picks for each user
New users	Struggles, shows popular items	Also needs some data, but learns quickly once they rate items

Conclusion

- The E-Commerce Recommendation Pipeline provides a structured approach for generating personalized product recommendations.
- The system collects user and product activity data, cleans and preprocesses the information, analyzes customer interests, identifies relevant products, and ranks them according to user preferences and product relevance.
- The project demonstrates how data processing and recommendation techniques can be used to improve product discovery, personalization, and customer experience in e-commerce platforms.

References

- **Adomavicius & Tuzhilin (2005):** Surveyed recommender systems and their evolution.
- **Linden et al. (2003):** Introduced Amazon's item-to-item recommendation approach.
- **Hu et al. (2008):** Focused on recommendations using implicit user feedback.
- **Koren et al. (2009):** Explained matrix factorization for personalized recommendations.
- **Covington et al. (2016):** Applied deep learning to YouTube recommendations.
- **He et al. (2017):** Proposed neural collaborative filtering for better recommendations.

THANK YOU